Project 1: Comparison of Two Classifiers in Predicting Turkish Noun, Pronoun, and Verbal Noun Cases

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1 Introduction

This paper investigates whether n-gram features can be used to train and classify Turkish nouns, pronouns, and verbal nouns by case effectively, determining which of two models is more effective for this task, and which cases are most and least accurately predicted. The Logistic Regression classifier, which is a linear classifier, and the Random Forest classifier, which is a non-linear classifier, are used. Additionally, the more effective model is tested on another annotated corpus, consisting of data from a different genre to determine whether the model remains effective across corpora.

Turkish is a language spoken primarily in Türkiye, and has around 90 million total speakers. It is an agglutinative language, meaning a large amount of information is encoded morphologically through the use of affixes concatenating with each other. Turkish, as a head-final language, exclusively uses suffixes for this purpose, including both derivational and inflectional suffixes.

Case is one of many suffix categories in Turkish, and is manifested in the language through standard suffixes which take into account vowel harmony, consonant harmony, and epenthesis (through the use of buffer letters. Turkish nouns are considered to have 6 main possible cases: the nominative case, the accusative case, the dative case, the locative case, the ablative case, and the genitive case. Additionally, as noted by Göksel and Kerslake (2005), the instrumental suffix, while not considered a case, has many case-like features, such as vowel harmony and epenthesis, and because marked in the dataset under the case column, it will be considered a case for the purposes of this paper. Because of the use of vowel and consonant harmony, each case can take many regular forms based on the preceding syllable. For example, the locative case can take the following forms: -da, -de, -ta, -te, -(n)da, -(n)de.

While this problem may seem simple, as each Turkish case suffix has a distinct pattern, there are many words, especially in the nominative (zero-marked), and dative (-a, -e, -(n)a, or -(n)e) cases, which may morphologically have an ending that mimics a suffix belonging to a different case without context.

In training the Logistic Regression and Random Forest classifiers to assess the reproducibility of learned features when assessing noun and verbal cases in Turkish, I hope to get a better understanding of whether a simple classifier could predict the case of the word with n-gram features and no additional context, and whether this classifier could successfully perform a similar task on an unrelated wordbank.

2 Methodology

This paper uses the BOUN Treebank, a corpus created by researchers at Boğazıçı University in Türkiye, composed of 9761 manually annotated Turkish sentences, with a total of 121,214 tokens, to train the two classifiers. The BOUN Treebank is a smaller part of the Turkish National Corpus, which contains 50 million words from a variety of sources. The sentences in the BOUN treebank specifically come from 5 of these types of sources: essays, newspapers, instructional texts, popular culture articles, and biographies (Türk et al., 2021). Each word is marked for features such as part of speech, lemma, form, case, number, person, tense, aspect, mood, evidentiality, and polarity, among other factors. To ensure accuracy, two native Turkish linguists labeled the treebank, and two other linguists cross-checked the annotations for accuracy.

The corpus was pre-split into training, development, and testing folders, with an 80%/10%/10% split. For this experiment, only the training and testing folders were downloaded and used.

Using the annotations provided in the dataset, words were filtered by their part of speech. Only nouns, pronouns, and verbal nouns were kept (which in the dataset, were given the "verb" category, as they retain many features of verbs in Turkish). Nouns, pronouns, and verbal nouns all had case annotated, and the words along with the case labels were retained. Lemmas, number, person, and other features were not used for this experiment.

2-4 character n-grams from the words were derived and vectorized, with character n-grams being chosen as most case information is encoded through suffixes. Both the Logistic Regression classifier and the Random Forest classifier were used to train the words separately, and then used to predict the cases of words in the test set. Precision, recall, F1-score, and support for each case (7 cases in total) were derived from the performance of both classifiers on the testing dataset. Additionally, a confusion matrix was generated for each classifier, showing each models' predictions compared to the correct labels.

From the two classifiers, the stronger classifier was used to annotate the KeNet Treebank. The KeNet Treebank is much larger than the BOUN Treebank (that was used for training the classifier), and contains 18,700 manually translated sentences and 178,700 tokens (Bakay et al., 2021). The sentences are all example sentences from the dictionary of the Turkish Language Association. While the annotations of the KeNet Treebank analyze for the same features that the BOUN Treebank does, the dictionary example sentences in KeNet include examples from books, speeches, and poems, a fundamentally different corpus makeup than that of the BOUN Treebank that the classifier was trained on, and thus, extending the existing classifier to the KeNet Treebank can test whether the accuracy, precision, recall, and other metrics can extend to a corpus with a different makeup.

For this process, the existing Logistic Regression classifier that was trained on the BOUN Treebank was used to annotate words in the KeNet Treebank. Nouns, verbal nouns, and pronouns, were filtered from the rest of the words to annotate. The existing BOUN-trained vectorizer was used to transform the words and predict their corresponding cases. This process was done for the entire KeNet Treebank, with precision, accuracy, F1-score, and support derived for each case category and for determining overall classifier performance on the task.

3 Results

Both the Logistic Regression and Random Forest classifiers performed effectively, with a 0.86 and 0.84 accuracy overall, respectively. However, the Logistic Regression classifier was more effective than the Random Forest classifier in several categories, including accuracy, overall precision, and overall recall.

Case	Pre.	Rec.	F1	Support
Abl	0.87	0.99	0.93	204
Acc	0.46	0.94	0.62	367
Dat	0.91	0.91	0.91	469
Gen	0.90	0.97	0.93	397
Ins	0.82	0.98	0.89	129
Loc	0.95	0.98	0.97	477
Nom	0.96	0.78	0.86	2468
Accuracy			0.86	4511
Macro Avg	0.84	0.94	0.87	4511
Weighted Avg	0.90	0.86	0.87	4511

Table 1: Precision, recall, and F1-score of Turkish case predictions (Logistic Regression)

The Logistic Regression classifier had an overall accuracy of 0.86, with weighted averages of 0.90 precision and 0.86 recall. However, there was some variation of these metrics across the different cases. The cases with the highest precision in the corpus were the locative and nominative cases, with a precision of 0.95 and 0.96, respectively. On the other hand, the case with the lowest precision, and the only case with a precision below 0.85, was the accusative case, with a precision of 0.46.

Additionally, out of the seven cases studied in this experiment, the nominative case accounted for 2468 of the 4511 nouns, pronouns, and verbal nouns overall. However, there was no case with any fewer than 100 support, with the instrumental case (in reality a suffix with case-like properties, as described earlier) being the case with the lowest support, at 129 (2.8%). Recall scores were high for most cases, with all cases having a recall above 0.90 except the nominative case, with a recall of just 0.78. Four cases (the ablative, genitive, instrumental, and locative cases), had recall of over 0.95, with scores of 0.99, 0.97, 0.98, and 0.98, respectively. The F1 scores were also high for most of the classes, with the lowest F1 belonging to the accusative case, with an F1 score of 0.62, weighed

down by the low precision score of the case.

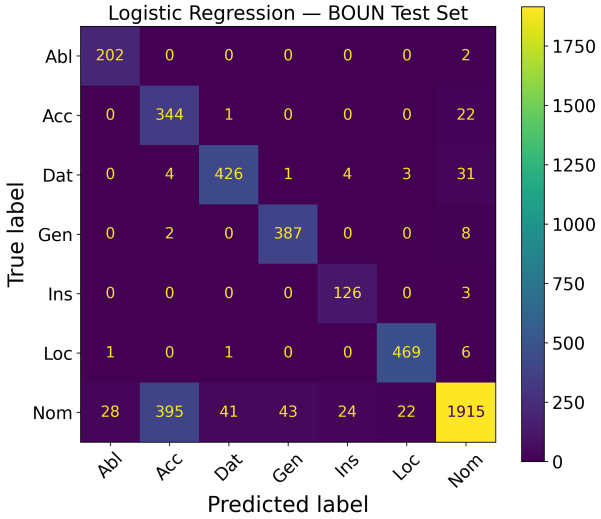


Figure 1: Confusion matrix for Turkish case predictions - BOUN Test Set (Logistic Classifier)

Figure 1 shows the confusion matrix for the Logistic Classifier for the BOUN test set. While most of the categories have a high degree of accuracy, with accurate case labels, the classifier often seemed to misclassify labels when the true label is the nominative case, with each case label being predicted at least 20 times when the true label was the nominative case. Additionally, the accusative case was often guessed in place of the nominative case. In fact, out of all the times the accusative case was guessed by the Logistic Regression classifier, the true label was the nominative case 395 times, even more than when the accusative case was the true value (344 times), the primary reason for the 0.46 precision in the accusative case.

Case	Pre.	Rec.	F1	Support
Abl	0.89	1.00	0.94	204
Acc	0.62	0.72	0.67	367
Dat	0.84	0.76	0.80	469
Gen	0.81	0.78	0.79	397
Ins	0.63	0.84	0.72	129
Loc	0.87	0.88	0.88	477
Nom	0.88	0.86	0.87	2468
Accuracy			0.84	4511
Macro Avg	0.79	0.83	0.81	4511
Weighted Avg	0.84	0.84	0.84	4511

Table 2: Precision, recall, and F1-score of Turkish case predictions (Random Forest)

Like the Logistic Regression classifier, the Random Forest classifier also had a strong accuracy score, although the overall precision and recall score were both 0.84, lower than the corresponding scores of the Logistic Regression classifier.

The case with the highest precision when using the Random Forest classifier was the ablative case, with a precision of 0.89. Notably, no case had a precision above 0.90, unlike the Logistic Regression classifier, which had 4 of these cases. On the other hand however, the case with the lowest precision was the accusative case, with a precision of 0.62. This was higher than the corresponding precision of 0.46 for the accusative case that the Logistic Regression classifier had. Notably, the accusative case was the case with the lowest precision for both classifiers. The Random Forest classifier also performed poorly with the instrumental suffix/case, with a precision of only 0.63, much lower than the corresponding 0.82 in the Logistic Regression classifier. Most of the recall scores of the Random Forest classifier were also lower than the corresponding scores of the Logistic Regression classifier. The lowest of these recall scores was that of the accusative case, with a recall of only 0.72, much lower than the corresponding score of 0.94 that the Logistic Regression classifier had. Additionally, the Random Forest classifier had recall scores below 0.80 for three cases (the accusative, dative, and genitive cases), contrasted with nominative case, which only the Logistic Regression classifier had a recall score of below 0.80 for. However, the Random Forest classifier had a perfect (1.00) recall with the ablative case, slightly above the also near-perfect score (0.99) the Logistic Regression classifier had.

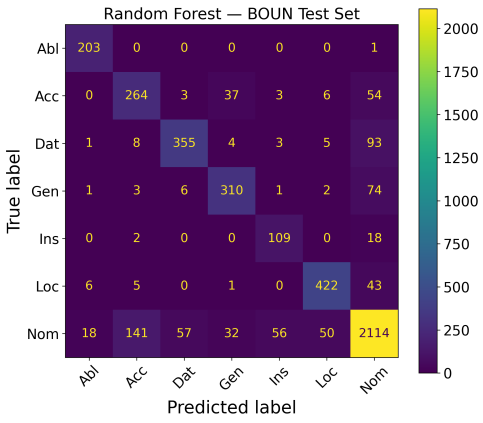


Figure 2: Confusion matrix for Turkish case predictions - BOUN Test Set (Random Forest)

Figure 2 shows the confusion matrix for the Random Forest classifier. As with the Logistic Regression classifier, the nomiative case is often mislabeled, especially when the predicted label is the accusative case, with the accusative case being labeled in place of the nominative case 141 times.

Case	Pre.	Rec.	F1	Support
Abl	0.91	1.00	0.95	3600
Acc	0.50	0.92	0.65	6980
Dat	0.86	0.91	0.88	8452
Gen	0.88	0.97	0.92	5792
Ins	0.79	0.95	0.86	1923
Loc	0.89	0.98	0.93	4743
Nom	0.96	0.75	0.84	38078
Accuracy			0.84	69568
Macro Avg	0.83	0.92	0.86	69568
Weighted Avg	0.88	0.84	0.85	69568

Table 3: Precision, recall, and F1-score of Turkish case predictions on the full KeNet corpus (Logistic Regression)

The Logistic Regression classifier was used to test with the KeNet corpus, as previously described, to determine whether correct case labels would be predicted on a corpus consisting of words from sources that were different than the data the classifier was trained on.

The classifier performed well on words from the KeNet corpus, achieving an overall accuracy of 0.84, just slightly below the accuracy of 0.86 that the Logistic Regression classifier achieved with the BOUN test data. The weighted average precision and recall were 0.88 and 0.84, respectively. Additionally, because the full KenEt corpus was used, the support for each case was quite large, although the nominative case still took up a majority of the data set (with 38,078 support out of the total of 69,568).

The classifier, when predicting labels for the KenEt corpus, as with the previous experiment, performed poorly on the accusative case, only achieving an precision of 0.50, very similar to its precision on the BOUN test dataset (0.46). However, most of the other cases had precision above 0.80, with the other exception being the instrumental suffix/case, with a precision of 0.79. The classifier also had exceptionally high recall, with a recall of above 0.90 on all of the case suffixes except the nominative case, which had a recall of 0.75. This was very similar to the performance of the Logistic Regression classifier on the BOUN test data, where it had a recall of 0.78. Overall, the performance of the classifier on the KenEt corpus, a new corpus it had not been previously exposed to, the classifier still performed strongly, and had strong scores for most of the cases, other than the low precision on the accusative case.

4 Discussion

The absolute strength of both models indicates that Turkish noun, verbal noun, and pronoun cases can often be predicted based on the word itself parsed with character n-grams, without additional context such as lemma and the surrounding words. This task was complicated by the fact that over half of the words (2468 words in the nominative case out of a total of 4511 words) analyzed in the testing data were in the nominative case. While Turkish is a pro-drop language and often drops subjects, many sentences still have an explicit subject, especially given Subjects in a Turkish sentence always take the nominative case (although this is not necessarily true for non-sentence clauses, such as embedded clauses).

Across most metrics, including accuracy, weighted average precision, and recall, the Logistic Regression classifier outperformed the Random Forest classifier. An explanation for the Logistic Regression classifier's stronger performance for this task as compared to Random Forest is due to the Logistic Regression classifier being a linear classifier, as opposed to Random Forest, which is a non-linear one. Due to this experiment using character n-grams for vectorizing words, these n-grams often acted as indicators of case suffixes at the end of a word, especially since many Turkish suffixes, including case suffixes are two or three characters long. Additionally, suffixes in Turkish follow regular patterns, and with very few irregularities, the classification boundary is quite linear. While Logistic Regression is very sensitive to outliers, Turkish's case suffix system has very few irregularities. Random Forest, on the other hand, uses feature subsampling to determine a category, which can often overfit weaker features in the data, causing a slightly lower accuracy as compared to a linear model. Turkish case, as mentioned previously, is exclusively marked at the end of the word, which makes word-ending n-grams crucial for predicting the case of a word, while n-grams

at the beginning of a word are not relevant. A Random Forest model, because of the use random resampling, may overfit word-leading n-grams, or not effectively sample word-ending case indicators, thus being slightly less effective than the Logistic Regression classifier.

Thus, differences in the performances of the two classifiers could often be understood through the lens of the differences between the classifiers themselves, along with an understanding of the dynamics of the suffixes linguistically. For instance, the Logistic Regression classifier boasted higher precision and recall scores across many of the specific cases, while the Random Forest classifier did not have a precision greater than 0.90 in any case. This is likely due to the idea of the Random Forest classifier occasionally randomly sampling and overfitting many character n-grams that contained little to no information about case, such as n-grams at the front of the word. On the other hand, the Random Forest classifier also had a strong performance specifically among the ablative case, with a precision of 0.89 and a recall of 1.00. This may be due to the ablative case (-dan, -den, -tan, or -ten) being the only case among the seven cases to be 3 letters long, even before any consonant insertion. Thus, this suffix is more likely to be included in many different character n-grams within a word, making the suffix more likely to be sampled at several distinct points by a non-linear classifier like the Random Forest classifier.

Another trend observed in the data is a poor performance among the accusative case. This was observed in both classifiers, along with the extension of the Logistic Regression classifier on to the KeNet corpus. A major reason for this due to the fact that in Turkish, the accusative case of a noun takes the same form as the 3rd person possessive (taken by the possessee, rather than the possessor) suffix when the word ends in a consonant. Both suffixes manifest as an -i, -ı, ü, or u depending on the vowel of the previous syllable, meaning that without context, it is not possible to determine whether a word of this type has the accusative suffix or the 3rd person singular possessive suffix. Words that take the third person singular possessive suffix, typically belong to the nominative case, unless additional case suffixes have been added to it. This is consistent with the confusion matrices, which showed a high amount of guesses of the accusative case for what was in reality the nominative case, and shows a problem in Turkish case prediction that cannot be solved without context.

Despite poor performance on the accusative case, the Logistic Regression performed well on the task of predicting case labels for the KeNet corpus, achieving a 0.84 accuracy overall, very similar to the 0.86 accuracy from the Logistic Regression classifier on the BOUN test dataset, even though the sentences and sources of the two datasets were very different. The explanation for this is that because a character n-gram model was used for annotating the case suffixes, the classifier was trained on the characters that composed the suffixes themselves, rather than word level information. Additionally, the Turkish suffix system and case suffix system is very regular, and can be extended to every noun in the language. Thus, even if the word usage in the KeNet corpus was different, the underlying suffix patterns and n-gram associations remained the same, and could be generalized to the new dataset.

Overall, both classifiers performed quite well in applying Turkish case suffixes to nouns, pronouns, and verbal nouns, even with no additional context. The Logistic Regression model performed better than the Random Forest model across most of the metrics, indicating that a linear approach is well-suited for the task of Turkish case prediction. The accusative case had the lowest precision for all three experiments, and the ablative case had the highest recall for all three experiments, alluding to linguistic details of the Turkish suffix system. These patterns all show that classifier performance is very much related to the structural properties of an individual language, and how performance in Turkish specifically is very dependent on the agglutinative system that the language relies on.

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